

CPSC 304 Project Cover Page

Milestone #: 1

Date: Feb 3, 2026

Group Number: 45

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Description:

a. Domain of the Application:

The domain of our application is healthy meal planning with a focus on nutrition. It reflects how people plan their meals in real life by considering calories, nutrients, dietary preferences, and food allergies. Many individuals and families want to eat healthier but find it difficult to keep track of nutritional information and ensure that meals meet their specific needs.

This application is different from a typical recipe or food ordering system. Instead of only listing recipes, it focuses on ingredients and their nutritional content, as well as how meals fit different health categories such as vegan or children's meals. By including user preferences and dietitian-designed recipes, the system supports more practical and informed meal planning decisions.

b. Aspects of the Domain Modeled by the Database:

Real-World Healthy Meal Planning Needs and Constraints

- Existing platforms treat “healthy meals” as a single category, ignoring differences between dietary goals.
- Many “healthy” recipes on existing platforms are based on informal or subjective interpretations of healthy eating, making it difficult for users to assess the reliability and authority of recipe information.
- Nutritional information is often incomplete or loosely structured.
- Allergy information typically requires manual checking at the ingredient level.
- Ingredient substitution is rarely supported in a structured manner.
- Users often need to evaluate meals at a set level rather than individual recipes.

How the Database Models and Supports These Needs

To address the challenges described above, our database models health-related information in a structured way, allowing recipes, ingredients, and dietary constraints to be represented and combined more clearly.

- Recipes are classified using **HealthMealType** (e.g., keto, vegetarian, plant-based), with an **ISA hierarchy** that supports overlapping diet categories such as **Vegan** and **ChildrenMeal**.
- Recipes in the system are designed by **Dietitians** with relevant certifications, which helps ensure that the recipes are authoritative, accurate, and based on scientific knowledge rather than informal interpretations of healthy eating.

- Each **Ingredient** is associated with detailed **Nutrient** information, supporting reasoning about overall nutrient intake and daily dietary limits.
- Allergen information is stored at the **Ingredient** level to support allergy-safe recipe filtering.
- **Ingredient** substitution is explicitly modeled through substitution relationships, enabling structured lookup of alternative ingredients.
- Recipes can be grouped into **MealSets**, allowing users to evaluate meals as a whole rather than as individual dishes.
- **Beverages** are modeled as a specific type of **Recipe** using an ISA relationship, allowing drinks to share common properties with other recipes while still supporting beverage-specific attributes.

In a real-life scenario, the application could be used by families planning meals for children with food allergies. For example, a parent preparing weekly meals for a child with peanut or egg allergies can use the system to exclude unsafe ingredients and select recipes that are classified as suitable for children. This helps ensure that meals meet both nutritional needs and safety requirements without manually checking each ingredient.

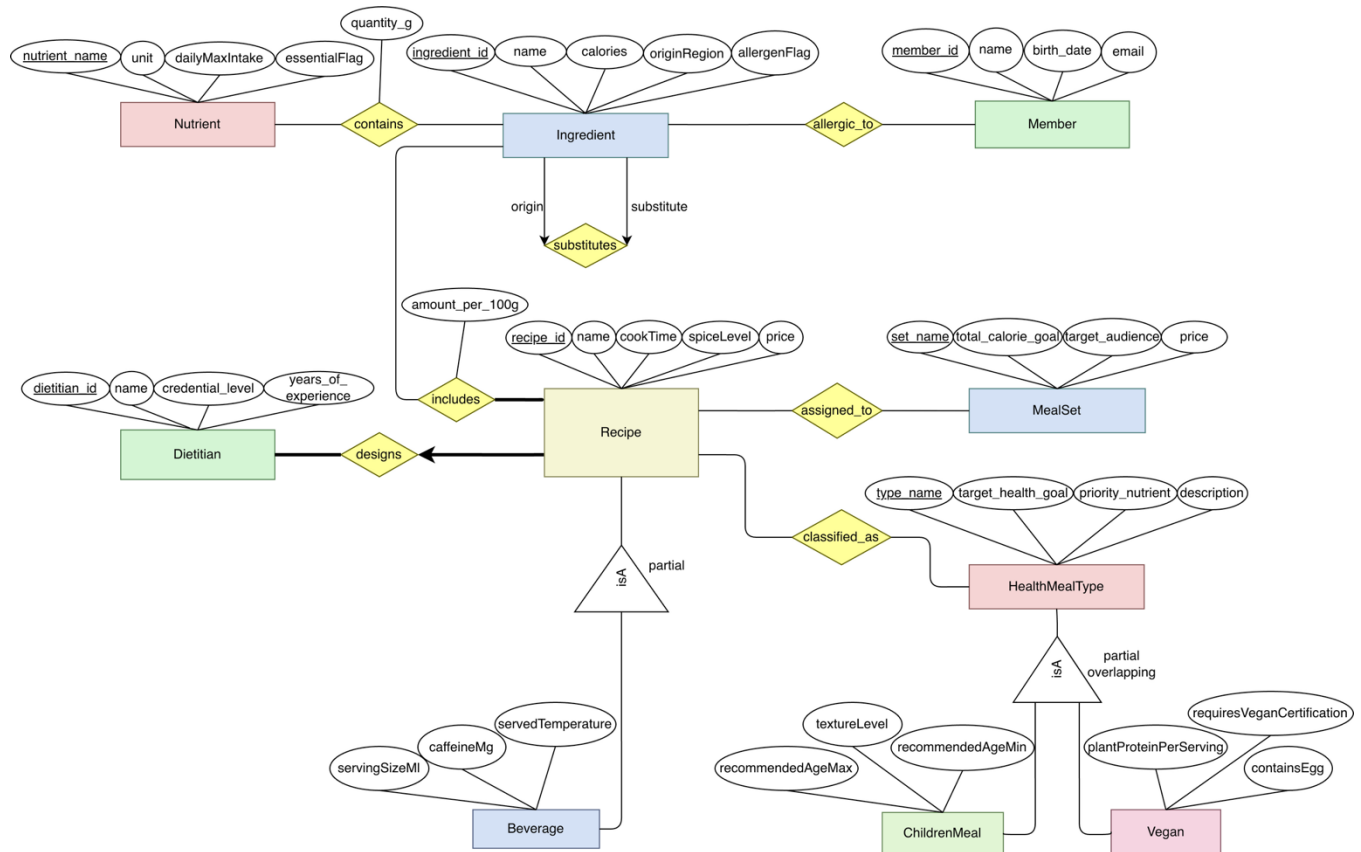
The application could also be used by individuals with specific health or dietary goals. For instance, an adult following a vegan diet or aiming to manage calorie intake can choose recipes based on dietary classification and nutritional content. By selecting meals designed by dietitians and aligned with personal preferences, the user can plan balanced meals that support long-term healthy eating habits.

Database Functionality

- **Store and manage recipe information**
The database stores core recipe information, including cooking time, spice level, price, and associations with ingredients and diet categories.
- **Allow dietitians to add new recipes using existing ingredients**
The database supports associating newly created recipes with existing ingredients in the system and recording the dietitian who designed each recipe.
- **Filter recipes by health meal type (ISA hierarchy)**
The database supports filtering recipes by health meal categories, including ISA subtypes such as Vegan and ChildrenMeal.
- **Exclude allergen-containing recipes for a given member**
Given a member's allergy constraints, the database supports filtering out recipes that contain allergenic ingredients.
- **Retrieve ingredients, quantities, and nutrient information for a recipe**
For a given recipe, the database supports retrieving its ingredients, their quantities, and associated nutrient information.

- **Compute total cooking time for a meal set**
The database supports aggregating cooking times of all recipes within a meal set to compute the total cooking time for the entire meal.
- **Support ingredient substitution**
The database supports querying possible substitute ingredients when a specific ingredient is unavailable or unsuitable.

ER Diagram:



Acknowledgment About Use of AI:

Yes, we used AI (ChatGPT) to suggest some cooking websites that provide recipes. We visited those websites for brainstorming ideas.

Please recommend some cooking websites that provide recipes.

Show thinking ▾

Whether you're looking for a quick weeknight dinner, a deep dive into the science of a perfect steak, or the secret to a flawless sourdough, there is a perfect site for you.

Here are my top recommendations categorized by what they do best:

The Gold Standards (Highly Reliable)

These sites are famous for testing their recipes dozens (sometimes hundreds) of times to ensure they work for anyone.

- [America's Test Kitchen](#): The ultimate authority on technique. They explain *why* a recipe works and which equipment is best. (Note: Most content requires a subscription, but it's the most trusted by pros).
- [Serious Eats](#): Best for food "nerds." It was founded on food science (largely by J. Kenji López-Alt) and provides incredibly detailed guides on everything from "The Food Lab" to regional world cuisines.
- [NYT Cooking](#): Known for elegant, high-quality recipes from top chefs. The comment section is famously helpful for finding tweaks and substitutions.

Best for Easy & Budget-Friendly Meals

- [Budget Bytes](#): Fantastic for students or anyone watching their wallet. It breaks down the cost per serving for every recipe and focuses on simple, delicious pantry staples.
 - [Pinch of Yum](#): Features very approachable, "real-life" recipes. They excel at "healthy-ish" comfort food and have a great section for Instant Pot and slow cooker meals.
 - [The Modern Proper](#): Specializes in "simple dinners for every day." Their recipes are modern, beautiful, and usually take under 45 minutes.
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